Essential in MX Road (40 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	Introduction to MX Road MXROAD key features Basic concepts **Getting started:** Create new project, Working environments CAD Environment View controls in MX CAD tools	Theory/Demo	2 Hrs
2	Session 2	Practice 01: Exercises included for practice	Practical	2 Hrs
3	Session 3	Enhanced menu and GUI changes **Survey inputs and validation:** General procedure (GENIO file) **String names and drawing styles:** Feature name sets Drawing produced by plan style sets Displaying the model, Hiding or deleting MX graphical elements **Survey input:** Google earth, (ASCII format) Excel	Theory/Demo	2 Hrs
4	Session 4	Practice 02: Exercises included for practice	Practical	2 Hrs
5	Session 5	**Status:** Deriving information from display - Point related Point selection methods Distance/ dimension related **Model analysis:** Triangulation Data conflicts Surface checker	Theory/Demo	2 Hrs
6	Session 6	Practice 03: Exercises included for practice	Practical	2 Hrs
7	Session 7	**Edit models:** Create model Delete a model Free/secure a model Rename a model Trim a model to a boundary Create composite model **Edit strings:** Create string dynamically Create string **Surface Analysis:** Performing surface analysis String Naming Convention	Theory/Demo	2 Hrs
8	Session 8	Practice 04: Exercises included for practice	Practical	2 Hrs
9	Session 9	**Design - quick alignment:** Alignment Horizontal design Horizontal design toolbar Report **Design -Vertical profile:** Design model Existing model Collinear profile model Vertical profile toolbar	Theory/Demo	2 Hrs
10	Session 10	Practice 05: Exercises included for practice	Practical	2 Hrs
11	Session 11	**Design- alignment by element method:** Begin horizontal design Horizontal alignment -Element method **Best fit alignment:** Best fit horizontal alignment- Element Best fit alignment- vertical string	Theory/Demo	2 Hrs
12	Session 12	Practice 06: Exercises included for practice	Practical	2 Hrs
13	Session 13	Road design- carriageways Rule based super elevation Cross fall checker Road design- road widening **Dynamic reports:**	Theory/Demo	2 Hrs

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		Types of reports		
		Adding annotation		
		Saving report		
		Exercises included for practice		
14	Session 14	Practice 07: Exercises included for practice	Practical	2 Hrs
15	Session 15	**Junction design:**	Theory/Demo	2 Hrs
		Create fillet		
		Regrade junction		
		Traffic island		
		Tidy junction		
		Road design- shoulder design		
		Road design- kerbs, verges and footways		
16	Session 16	Practice 08: Exercises included for practice	Practical	2 Hrs
17	Session 17	**Earthwork design:**	Theory/Demo	2 Hrs
		Earthwork model details		
		Assign Earthwork Style and Strategy		
		Earthwork style open		
		Alternative strategy		
		End earthworks		
18	Session 18	Practice 09: Exercises included for practice	Practical	2 Hrs
19	Session 19	**Pavement and subgrade design:**	Theory/Demo	2 Hrs
		Model details		
		Pavement layer design		
		Pavement layer design- Style set, Pavement styles –Areas and Zones, Reports		
		Final Drawings Manager:		
		Create cross sections		
		Display cross sections		
		Layout pages		
		Page setup		
		Create or edit page setup		
		Page layout		
		Edit page layout		
		Annotate drawing		
		Publish drawings		
20	Session 20	Practice 10: Exercises included for practice	Practical	2 Hrs